Finding the best feasible bundle

Lecture notes

What this lecture is about. Last lecture ended with a picture: the best feasible bundle is where the highest reachable indifference curve touches the budget line. Today we go slowly. First the utility functions, one form at a time. Then we look at four kinds of preferences on one budget and see that the best bundle can sit at a corner, at a kink, or where a curve just touches the line. Then we walk along the budget line to see *why* the touching point is best. Then we draw the same idea with arrows, and only then write it as algebra (the Lagrangian). We end with corners, kinks and three exercises.

One budget for §§2–6. Prices $p_x = 3$, $p_y = 1$, money $B = 60$:

$$3x + y = 60.$$

1 Utility functions, one form at a time

Utility function

A utility function gives every bundle a number, so that a better bundle gets a higher number and equally good bundles get the same number. An indifference curve is all bundles with the same number. Only the order of the numbers matters (ordinal): $2u + 5$ or u^2 describe the same preferences.

form	formula	indifference curves
perfect substitutes	$u = a\,x + b\,y$	straight lines; the MRS is a/b everywhere
perfect complements	$u = \min\{x, y\}$	L-shapes, kinks on the line $x = y$
Leontief (the general recipe)	$u = \min\{a\,x, b\,y\}$	L-shapes, kinks on the line $a\,x = b\,y$
Cobb-Douglas	$u = x\,y$ (or $x^\alpha y^\beta$)	smooth, bowed in
quasi-linear	$u = \alpha x + y^\beta$, e.g. $2x + \sqrt{y}$	each curve is the same curve shifted; bowed in for $\beta < 1$
quasi-linear, y squared	$u = x + y^2/60$	shifted copies, but bowed *out*

Perfect complements are the 1 : 1 Leontief ($a = b = 1$). Example of a recipe: every coffee needs exactly 2 sugars, $u = \min\{2x, y\}$, with kinks on $y = 2x$.

A number for every bundle. With $u = x\,y$: $(2, 8)$, $(4, 4)$ and $(8, 2)$ all get 16, so they lie on one indifference curve; $(4, 8)$ and $(8, 4)$ get 32, on a higher one. Double every number, $v = 2xy$: 16 becomes 32 and 32 becomes 64, but the curves do not move and the order stays. Only the order matters: utility is *ordinal*.

Same bundle, different preferences. On the parameter sliders every curve goes through the same bundle $A = (10, 10)$: each curve is a *different person*. One person's indifference curves never cross; different people's can. Ask each person: one more x, how much y would you give? That is the MRS at A:

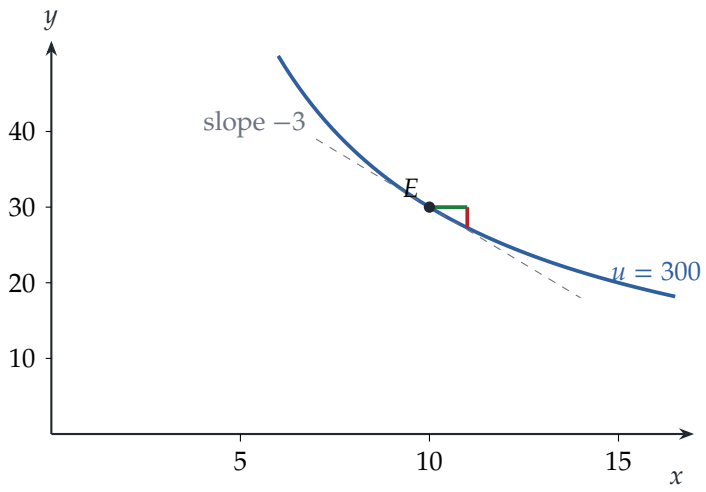
$$\text{substitutes: } \frac{a}{b}, \qquad \text{Cobb-Douglas: } \frac{\alpha}{\beta} \cdot \frac{y}{x}, \qquad \text{quasi-linear: } \frac{\alpha}{\beta} y^{1-\beta}.$$

For perfect complements: on the flat arm nothing (extra x alone adds nothing), on the steep arm the spare y down to the kink. A steeper curve through A means a person who likes x more.

Marginal utility and the MRS from one curve

Marginal utility. With $u = x\,y$ at $A = (5, 45)$: $u = 225$. One more x, the same y: $6 \cdot 45 = 270$, a gain of 45. Why 45? One more x adds one more "y": at $C = (15, 15)$, $16 \cdot 15 = 240$, a gain of $15 = y$. So $MU_x = y$, and in the same way $MU_y = x$ (at A: $5 \cdot 46 = 230$, a gain of 5).

The MRS by staying on one curve. Take the curve $u = x\,y = 300$; it passes through $E = (10, 30)$, and on it $y = 300/x$.



Step from $x = 10$ to 11, staying on the curve: $y = 300/11 = 27.27$, so $\Delta y/\Delta x = -2.727$. Smaller steps give -2.970 ($\Delta x = 0.1$) and -2.997 ($\Delta x = 0.01$), going to -3. With calculus:

$$\frac{dy}{dx} = -\frac{300}{x^2} = -\frac{y}{x} = -\frac{30}{10} = -3 \quad \text{at } E.$$

The general way, for any utility function: along the curve utility stays the same, so for small steps the gain from more x equals the loss from less y:

$$MU_x\,dx + MU_y\,dy = 0 \qquad \Longrightarrow \qquad MRS = -\frac{dy}{dx} = \frac{MU_x}{MU_y}.$$

Here $MRS = y/x = 30/10 = 3$: the same answer both ways. At $A = (5, 45)$: $MRS = 45/5 = 9$: I would give 9 y for one more x.

Quasi-linear, the other way: y squared

With $u = x + y^2/60$ (the $1/60$ only makes the picture visible), $MU_x = 1$ and $MU_y = y/30$, so $MRS = 30/y$: the curves are again shifted copies, but now they bow *out*, because more y makes y worth more, not less. On the budget $3x + y = 60$:

bundle	$(20, 0)$	$(50/3, 10)$	$(10, 30)$	$(0, 60)$
u	20	18.33	25	**60**

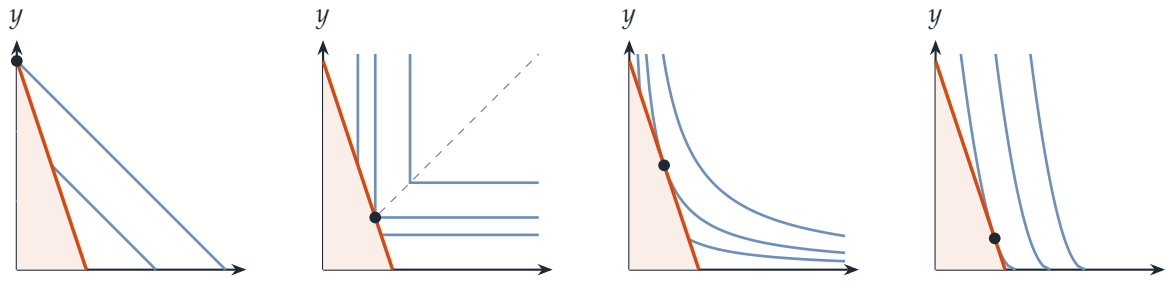
The curve just touches the line at $y = 10$ (where $30/y = 3$), and that point is the *worst* on the line. The best is the corner $(0, 60)$. Touching is not enough when the curves bow out.

Interior solution, corner solution

An *interior solution*: the best feasible bundle has some of both goods. With smooth curves it is where a curve just touches the budget line. A *corner solution*: all the money goes on one good. With perfect complements the best bundle is at a *kink*: interior (some of both), but not a touching point, because the curve has no single slope there.

2 Four kinds of preferences on one budget

Each panel shows indifference curves (blue) and the same budget line (orange). The black dot is the best feasible bundle.



perfect substitutes
$u = x + y$: a **corner**

perfect complements
$u = \min\{x, y\}$: a **kink**

Cobb-Douglas
$u = xy$: **just touches**

quasi-linear
$u = x + 2\sqrt{y}$: **just touches**

1. **Perfect substitutes,** $u = x + y$ (the same energy drink; x: at the U-Bahn kiosk, y: at Billa; $p_x = 3, p_y = 1$). A can is a can, so one for one: straight lines, MRS = 1. But the kiosk charges three times as much, so you buy them all at Billa: the **corner** $(0, 60)$, $u = 60$.

2. **Perfect complements (Leontief),** $u = \min\{x, y\}$ (hot dogs; x: sausages, y: buns, one each; $p_x = 3, p_y = 1$). A sausage without a bun adds nothing, so only pairs count, and the curves are L-shaped. Their **kinks** sit on the line $x = y$, where nothing is wasted. On the budget: $3x + x = 60$, so 15 hot dogs, $(15, 15)$.

3. **Cobb-Douglas,** $u = xy$ (a party; x: pizza slices, y: soda cans). I like both; the more x I already have compared with y, the less y I give up for one more x, so the curves bow in. A curve **just touches** the line at $(10, 30)$, $u = 300$. This is our point E for the rest of the notes.

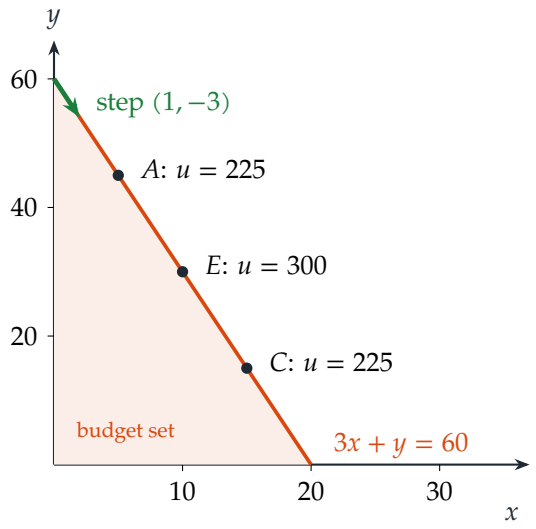
4. **Quasi-linear,** $u = x + 2\sqrt{y}$ (a student's month; x: Mensa lunches, y: coffees; $p_x = 3$, $p_y = 1$). Every lunch is worth the same; the first coffee wakes you up, the ninth does little. The MRS is $\sqrt{y}$: it depends on y only, so every curve is the same curve shifted sideways. It just touches where $\sqrt{y} = 3$: 9 coffees and 17 lunches, $(17, 9)$, $u = 23$.

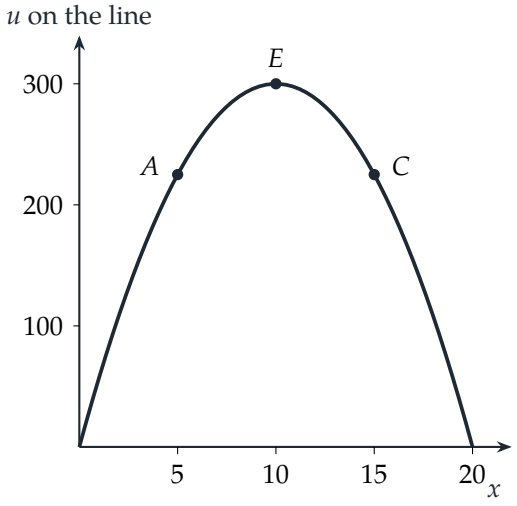
Insight: three shapes of the answer

The best feasible bundle is at a *corner*, at a *kink*, or where a curve *just touches* the line. The rest of today explains why the touching point is best (§§3–6), and why corners and kinks need a different check (§7).

3 Walk along the budget line

From now on until §7: $u = xy$ on $3x + y = 60$.



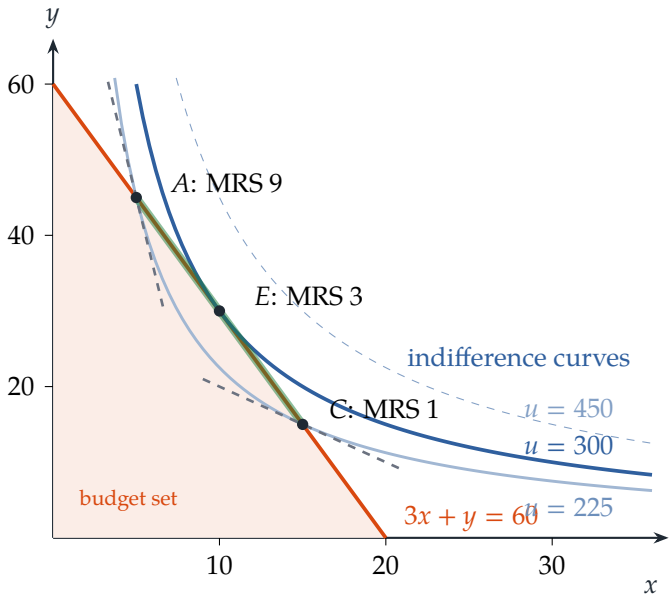


Start at $(0, 60)$: all money on y, but $u = 0 \cdot 60 = 0$. One step right is one more x and three less y; it costs $3 - 3 = 0$, so we stay on the line. At $A = (5, 45)$, $u = 225$. At $E = (10, 30)$, $u = 300$. At $C = (15, 15)$, $u = 225$ again. (C is the bundle the hot-dog person picked in §2; these preferences are different.) Along the line, utility is

$$u = x\,(60 - 3x),$$

which rises and then falls, like walking over a hill: $0, 225, 300, 225, 0$. **The top of the hill, E, is the best feasible bundle.** In 3D: utility is a hill over the map, and the budget line is a trail across it; walking the trail you go up and then down, and its highest point is E.

4 At the top, the curve just touches: the same slope



A and C have the same $u = 225$, so they lie on one indifference curve, and it *cuts* the line twice. Every bundle on the line between the two cuts (green) lies on a higher curve; that is the hill of §3, where utility on the line is above 225 between A and C. So neither A nor C can be best. At E the curve $xy = 300$ no longer cuts the line; it only touches it. Higher curves, like 450, miss the line: nothing on them is affordable.

Marginal utility

MU_x is how much u rises with one more unit of x, holding y fixed (a derivative). For $u = xy$: $MU_x = y$ and $MU_y = x$. Check at A: from $(5, 45)$ to $(6, 45)$, u goes from 225 to 270, up by $45 = y$.

The slope of the line is the price ratio, the slope of the curve is the MRS (grey dashes in the picture):

$$\frac{p_x}{p_y} = \frac{3}{1} = 3$$

$$MRS = \frac{MU_x}{MU_y} = \frac{y}{x}: \qquad MRS(A) = \frac{45}{5} = 9, \qquad MRS(C) = \frac{15}{15} = 1, \qquad MRS(E) = \frac{30}{10} = 3.$$

At A I would give 9 y for one more x, and the shop asks only 3: buy more x. At C I would give only 1, the shop asks 3: buy less x. At E my swap rate equals the shop's: no swap helps. **Touching = the same slope:** $MRS = p_x/p_y$.

5 Bang for the buck

Divide each marginal utility by its price: the extra utility from one more euro spent on that good.

bundle	MU_x/p_x	MU_y/p_y	so ...
$A = (5, 45)$	$45/3 = 15$	$5/1 = 5$	move euros to x
$C = (15, 15)$	$15/3 = 5$	$15/1 = 15$	move euros to y
$E = (10, 30)$	$30/3 = 10$	$10/1 = 10$	nothing to gain

At the best feasible bundle the last euro does equally well on both goods. **Remember this number, 10: it comes back as λ.** Equal bang for the buck, $MU_x/p_x = MU_y/p_y$, is the same condition as $MRS = p_x/p_y$ (multiply both sides by p_x/MU_y).

6 Arrows

6.1 An arrow written as two numbers

Vector and dot product

A *vector* is an arrow written as two numbers: how far right, how far up. $(3, 1)$ is 3 right and 1 up; $d = (1, -3)$ is 1 right and 3 down. Only its direction and length matter, not where we draw it.

The *dot product* multiplies the matching numbers and adds them:

$$(a_1, a_2) \cdot (b_1, b_2) = a_1 b_1 + a_2 b_2.$$

It is 0 exactly when the two arrows stand at a right angle.

6.2 The price arrow stands at a right angle to the budget line

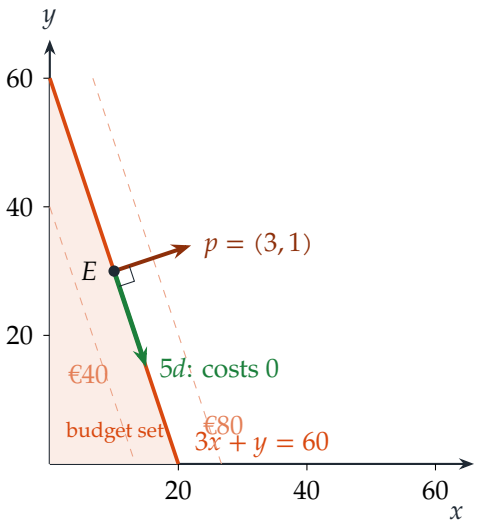
Put the prices into an arrow, $p = (3, 1)$. Dotted with any step (dx, dy), it gives exactly what that step costs:

$$p \cdot (dx, dy) = 3\,dx + 1\,dy.$$

A step along the budget line, $d = (1, -3)$, costs

$$p \cdot d = 3 \cdot 1 + 1 \cdot (-3) = 0,$$

so p stands at a right angle to the line.



Why p is also the fastest way to spend more: the same-spending lines $(3x + y = 40, 60, 80)$ are parallel, and the shortest way from one to the next goes straight across, at a right angle, which is the direction of p. In 3D: spending is a flat roof over the map, the budget line is its €60 height line, and p points straight up the roof.

6.3 The utility arrow: the steepest way up the mountain

Utility is a mountain over the map; the indifference curves are its height lines, like on a hiking map (Frank, App. 3A, Fig. 3A.3). Put the two marginal utilities into an arrow, the *gradient*:

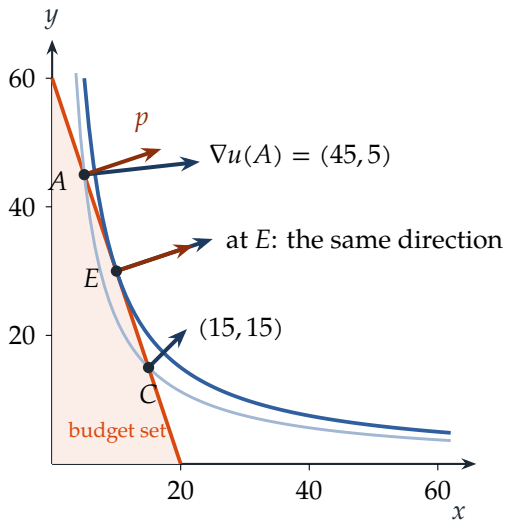
$$\nabla u = (MU_x, MU_y) = (y, x).$$

At A it is $(45, 5)$: one more x adds exactly 45, one more y exactly 5. So a small step changes utility by

$$du = 45\,dx + 5\,dy = \nabla u \cdot (dx, dy),$$

the same shape as its cost $p \cdot (dx, dy)$, with marginal utilities in place of prices.

Why the gradient stands at a right angle to the indifference curve. Along the curve u does not change, so a small step along the curve has $\nabla u \cdot \text{step} = 0$: a right angle, for the same reason as p and the budget line. And as for p: the shortest way from one height line to the next goes straight across, so the gradient is the steepest way up.



Insight: two arrows, two right angles

p stands at a right angle to the same-spending line; ∇u stands at a right angle to the same-utility curve. At A they point different ways, so the curve and the line cross: A is not best. At E they point the same way, so the curve and the line touch: E is best.

6.4 Split the arrow: the piece along the budget line and the piece along p

I must stay on the budget line, so only the part of the arrow along the line can help. One step d from A gains 45 from the extra x and loses $3 \cdot 5 = 15$ from the three fewer y:

$$\nabla u \cdot d = 45 \cdot 1 + 5 \cdot (-3) = 30.$$

For small steps this is the change in utility per step. Check with shrinking steps from A: the gain per unit of step is 27 (step 1: $u(6, 42) = 252$), 29.7 (step 0.1) and 29.97 (step 0.01), going to 30. Its *sign* tells which way along the line is better.

Now split the arrow in two pieces, one along the line ($a\,d$) and one along p ($m\,p$):

$$\nabla u = a\,d + m\,p.$$

Because $p \cdot d = 0$, the split is easy: dot both sides with d, then with p (and $d \cdot d = p \cdot p = 10$):

$$a = \frac{\nabla u \cdot d}{d \cdot d}, \qquad m = \frac{\nabla u \cdot p}{p \cdot p}.$$

point	∇u	along the line $a\,d$	along p: $m\,p$	so ...
A	$(45, 5)$	$3d = (3, -9)$	$14p = (42, 14)$	walk toward more x
C	$(15, 15)$	$-3d = (-3, 9)$	$6p = (18, 6)$	walk back, toward less x
E	$(30, 10)$	0	$10p = (30, 10)$	nothing left to gain

The piece along p is a move we can't make without more money. At E there is no piece along the line. Walking t steps from A,

$$u(5+t,\ 45-3t) = 225 + 30t - 3t^2,$$

which is highest at $t = 5$: the point $(10, 30) = E$, the top of the hill of §3.

6.5 At E, the utility arrow is the price arrow stretched

At E the whole arrow is the piece along p. Same direction, different length: stretch p until it covers ∇u. It fits at 10:

$$(30, 10) = 10 \cdot (3, 1), \qquad \text{that is} \qquad \nabla u = \lambda\, p \quad \text{with} \quad \lambda = 10.$$

It had to fit: p stands at a right angle to the budget line, ∇u at a right angle to the indifference curve, and at E the two touch. At A no stretch fits: $45/3 = 15$ but $5/1 = 5$.

λ is the value of one more euro: €1 more on x gives $(10 + \frac{1}{3}) \cdot 30 - 300 = 10$; €1 more on y gives $10 \cdot 31 - 300 = 10$. The same 10 as the equal bang for the buck.

Insight: three pictures, one condition

$$\frac{MU_x}{MU_y} = \frac{p_x}{p_y} \qquad \Longleftrightarrow \qquad \frac{MU_x}{p_x} = \frac{MU_y}{p_y} \qquad \Longleftrightarrow \qquad \nabla u = \lambda\, p.$$

Touching curves, equal bars, parallel arrows: divide the arrow's two numbers and you get the other two.

7 The Lagrangian: the same picture in algebra

We need three things at once: the two numbers of $\nabla u = \lambda p$, and the budget line. λ is a new unknown, solved for together with x and y; we already know it is the utility value of one euro. Lagrange packs all three into one formula, utility plus λ times the money left over:

$$\mathcal{L} = x\,y + \lambda\,(60 - 3x - y).$$

Set each slope to zero, one at a time:

$$\frac{\partial \mathcal{L}}{\partial x} = y - 3\lambda = 0 \qquad (MU_x = \lambda\, p_x)$$

$$\frac{\partial \mathcal{L}}{\partial y} = x - \lambda = 0 \qquad (MU_y = \lambda\, p_y)$$

$$\frac{\partial \mathcal{L}}{\partial \lambda} = 60 - 3x - y = 0 \qquad (\text{the budget line})$$

Solve, one line at a time:

$$x = \lambda$$

$$y = 3\lambda \qquad (\text{so } y = 3x\text{: MRS} = 3 = \text{the price ratio})$$

$$3\lambda + 3\lambda = 60 \quad \Longrightarrow \quad \lambda = 10$$

$$(x, y) = (10, 30)$$

The same E, and the same stretch factor as the arrows.

What λ measures, with numbers. With any money B the same steps give $x = B/6$ and $y = B/2$, so the best utility is

$$u^* = \frac{B}{6} \cdot \frac{B}{2} = \frac{B^2}{12}.$$

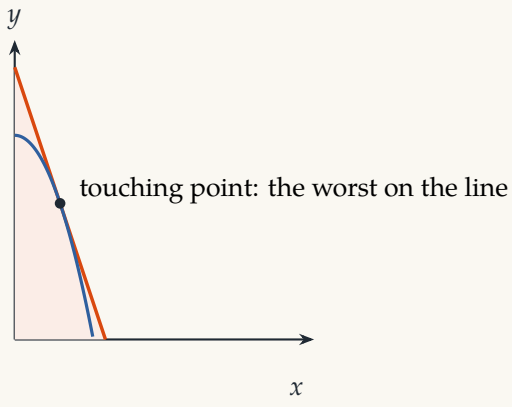
money B	u^*	gain per euro
60	300	
61	310.0833	10.0833
60.1	301.000833	10.00833
60.01	300.1000083	10.00083

As the extra money shrinks, the gain per euro goes to $10 = \lambda$. (With $u = 2xy$ the arrow doubles and $\lambda = 20$, but E stays: the number depends on how we number utility, its meaning does not.)

More money: the income-consumption curve. With any money B the best bundle is $x = B/6$, $y = B/2$, so as B changes the best bundles move along the line $y = 3x$. Frank calls this path the *income-consumption curve* (p. 98, Fig. 4.3). In 3D: each budget is its own trail on the utility hill, and the tops of the trails lie on $y = 3x$.

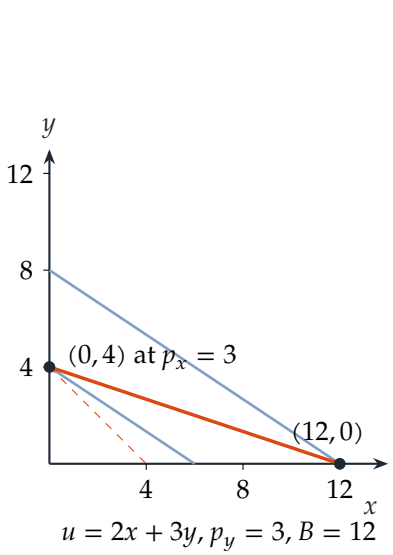
Insight: why the touching point is a best point

It works because the curves bow in. On a curve that bowed out, the touching point would be the *worst* bundle on the line: the two ends of the line would lie on higher curves.

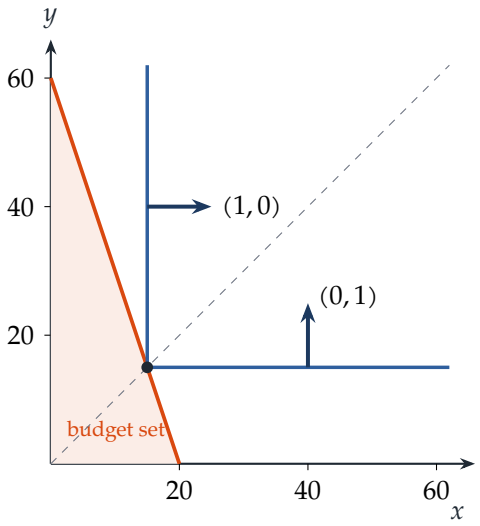


8 Corners and kinks

The rule $\nabla u = \lambda p$ (or $MRS = p_x/p_y$) is for a best bundle on the line away from both ends, where the curve is smooth. Two cases need a different check.



$u = 2x + 3y$, $p_y = 3$, $B = 12$



$u = \min\{x, y\}$ on $3x + y = 60$

A corner: perfect substitutes, $u = 2x + 3y$, with $p_y = 3$ and $B = 12$. The utility arrow is $(2, 3)$ at every point. With $p_x = 1$ the step along the budget line is $d = (3, -1)$, and gain minus loss is

$$\nabla u \cdot d = 2 \cdot 3 - 3 \cdot 1 = 3 > 0$$

at every point. So we walk toward x until the line ends at $(12, 0)$, and would like to walk on. At a corner ∇u is not a multiple of p, and that is fine. With $p_x = 3$ it flips: $(0, 4)$. With $p_x = 2$, gain minus loss is 0 everywhere: the whole line is best.

A kink: perfect complements, $u = \min\{x, y\}$ on $3x + y = 60$. Above the kink only more x helps (arrow $(1, 0)$); to the right of it only more y helps (arrow $(0, 1)$). At $(15, 15)$ the two arrows disagree, so there is no single steepest way up. Walking along the line either way makes fewer pairs: $(16, 12)$ gives 12 and $(14, 18)$ gives 14, against 15 at the kink. The check here is "no waste": buy on the ray $x = y$.

Interior or corner: quasi-linear, $u = 2x + \sqrt{y}$ (L08's mixture), $p_x = 8$, $p_y = 1$. Here

$$MRS = \frac{2}{1/(2\sqrt{y})} = 4\sqrt{y}.$$

With $B = 20$: $4\sqrt{y} = 8$ gives $y = 4$, and the remaining 16 buy $x = 2$: the bundle $(2, 4)$. With $B = 2$ we cannot even pay for $y = 4$, so everything goes on y: the corner $(0, 2)$, where $4\sqrt{2} < 8$; we would like less x, but x is already 0. So: $y = 4$ whatever the money, *once there is at least €4*; beyond that, extra money goes only into x. As money grows, the best bundle first climbs the y-axis up to $y = 4$, then moves straight right.

9 Exercises

Exercise 1 · climbing and cinema (Cobb-Douglas)

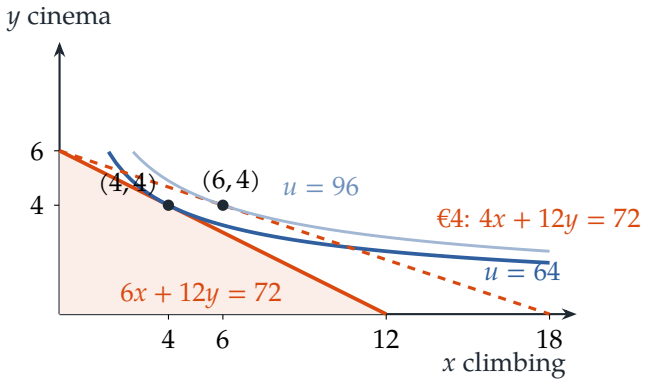
$p_x = 6,\ p_y = 12,\ B = 72$; x: climbing-gym visits, y: cinema tickets; $u = x\,y^2$.

1. The budget line: $6x + 12y = 72$, from $(12, 0)$ to $(0, 6)$.
2. Walk along it: at $x = 0, 2, 4, 6, 8, 10, 12$, $u = 0, 50, 64, 54, 32, 10, 0$. The top is at $(4, 4)$.
3. The curve $u = 64$ just touches the line at $(4, 4)$; $u = 50$ cuts it twice; $u = 100$ misses it.
4. Touching means the same slope:

$$MRS = \frac{MU_x}{MU_y} = \frac{y^2}{2xy} = \frac{y}{2x}$$

$$\frac{y}{2x} = \frac{6}{12} \quad \Longrightarrow \quad y = x$$

5. Into the budget: $6x + 12x = 72$, so $x = 4$ and $(x, y) = (4, 4)$. Check: $24 + 48 = 72$.
6. The arrows: $\nabla u(4, 4) = (16, 32) = \frac{8}{3} \cdot (6, 12)$, so $\lambda = 8/3$.
7. **Twist:** p_x falls from 6 to 4. Then $y/(2x) = 4/12$, so $y = 2x/3$, and cinema spending $12y = 8x$ is twice climbing spending $4x$: cinema gets $2/3$ of €72 = €48 = 4 tickets. $4x + 8x = 72$ gives $(6, 4)$: the same 4 tickets, more climbing.



Exercise 2 · swims and everything else (quasi-linear)

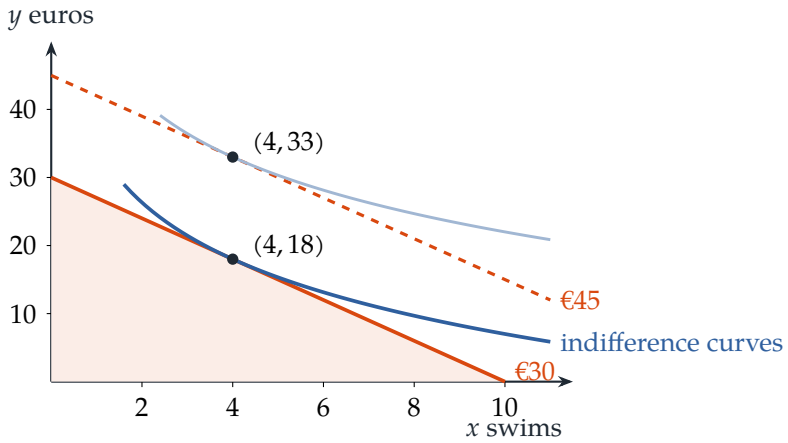
$p_x = 3,\ p_y = 1,\ B = 30$; x: swims, y: euros for everything else; $u = 12 \ln x + y$.

1. The budget line: $3x + y = 30$.
2. What one more swim is worth (the derivative of $\ln x$ is $1/x$): $MU_x = 12/x$, $MU_y = 1$.
3. A swim costs €3 of everything else, and each euro of y is worth exactly 1, so stop where

$$\frac{12}{x} = 3 \quad \Longrightarrow \quad x = 4, \qquad y = 30 - 12 = 18.$$

4. More money ($B = 45$): the condition $12/x = 3$ has no money in it, so still 4 swims, and $y = 33$. The best point moves straight up.

5. **Twist:** $B = 9$. Four swims would cost €12. Even at 3 swims, $12/3 = 4 > 3$: a swim is still worth more than it costs, so every euro goes to swims: the corner $(3, 0)$. Below €12 you spend everything on swims; from €12 up you buy 4 and keep the rest.



Exercise 3 · bracelets (perfect complements)

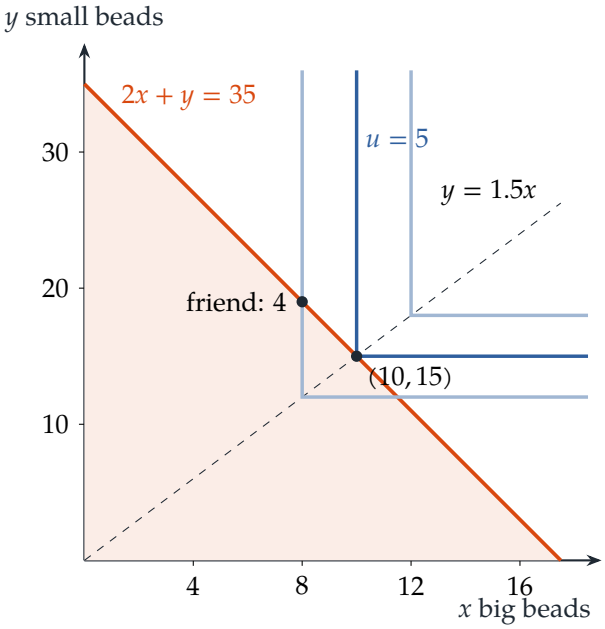
$p_x = 2$, $p_y = 1$, $B = 35$; x: big beads, y: small beads; a bracelet = 2 big + 3 small; $u = \min\{x/2,\ y/3\}$.

1. The recipe ray: $y = \frac{3}{2}x$. The indifference curves are L-shapes with kinks on the ray: extra beads of one kind alone make no new bracelet.

2. One bracelet costs $2 \cdot 2 + 3 \cdot 1 = 7$ euros, so €35 buys $35/7 = 5$ bracelets: $(x, y) = (10, 15)$. Check: $20 + 15 = 35$.

3. **Twist:** a friend spends €35 on 8 big and 19 small beads: $\min\{8/2,\ 19/3\} = 4$ bracelets. The other 7 small beads are wasted.

4. Why not "MRS = price ratio": at the kink the L has no single slope. Find the ray, price one set, divide the budget.



What to remember

1. The best feasible bundle is at a corner, at a kink, or where a curve just touches the budget line.
2. Touching: *MRS* = price ratio = equal bang for the buck = the utility arrow is λ times the price arrow.
3. The Lagrangian writes the same picture in algebra; λ = the utility from one more euro.
4. Corners and kinks: walk to the end of the line, or buy in the fixed ratio.

Reading. Frank, *Microeconomics and Behavior*, pp. 71–76 and Appendix 3A, pp. 87–92 (the utility mountain: Fig. 3A.3). Online: M. J. Osborne, *Mathematical Methods for Economic Theory* (free tutorial), §2.3 (the gradient stands at a right angle to the level curve) and §6.1 (one constraint; what λ means), `https://mjo.osborne.economics.utoronto.ca/index.php/tutorial`.